

“**NATURE** IS ALL THAT A MAN BRINGS WITH HIMSELF **INTO THE WORLD**; **NURTURE** IS EVERY INFLUENCE WITHOUT THAT AFFECTS HIM **AFTER HIS BIRTH.**”

Francis Galton, from *English Men of Science: Their Nature and Nurture*, 1874

GERMAN MATHEMATICIAN

GEORG CANTOR (1845–1918) published *On a Characteristic Property of all Real Algebraic Numbers* in 1874. This paper laid the foundations of **set theory**, and introduced the idea of different kinds of **infinity**.

The same year, Russian mathematician **Sofia Vasilyevna Kovalevskaya** (1850–91) became the **first woman to be awarded a doctorate in mathematics**, which she received from the University of Göttingen, Germany. In 1889, she became the **first woman to be appointed professor of mathematics**, at Stockholm University, Sweden.

Meanwhile, British economist **William Stanley Jevons** (1835–82) published *The Principles of*

Science: A Treatise on Logic and Scientific Method, which criticized the method of **induction** as a source of new scientific ideas and instead recommended **random hypotheses**.

Austrian chemist Othmar Zeidler (1849–1911) synthesized **DDT** (dichlorodiphenyltrichloroethane) as a doctoral student at the University of Strasbourg, France, in 1874. The importance of the substance as a powerful insecticide was not realized until the late 1930s.

French chemist **Paul-Émile Lecoq de Boisbaudran** (1838–1912) plugged a gap in Mendeleyev's periodic table of elements (see 1868–69) by identifying the metal **gallium** spectroscopically in 1875. Before

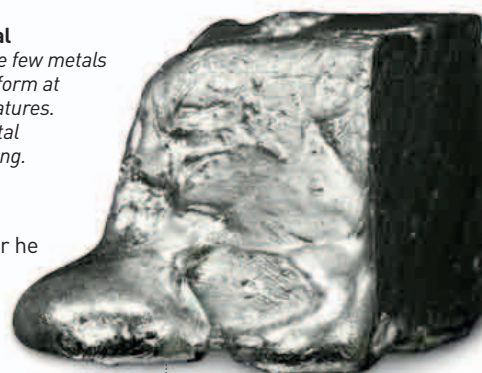
Silvery white metal

Gallium is one of the few metals that occur in liquid form at near-room temperatures. Like water, this metal expands on solidifying.

the end of the year he had synthesized pure gallium by the electrolysis of a solution of gallium hydroxide in potassium hydroxide.

Austrian geologist **Eduard Suess** (1831–1914) coined the term **biosphere** (see panel, left) in 1875, defining it as “the place on Earth's surface where life dwells.” The concept was intended to complement the **three geological zones**—the lithosphere (the rocky outer layer of the planet), the hydrosphere (the layer of water at Earth's surface), and the atmosphere (the gas envelope surrounding Earth). However, the theory made little impact on the scientific community until it was developed by Russian geochemist Vladimir Vernadsky (1863–1945) in his 1926 book, *La Biosphere*.

The scientific study of twins began in 1875, when British scientist **Francis Galton** (1822–1911) published his landmark paper *The History of Twins, as a Criterion of the Relative Powers of Nature and*



Nurture. While he paved the way for further scientific enquiry into this subject, Galton was unaware of the distinction between monozygotic (identical) twins, born from one fertilized egg, and dizygotic twins born from two. These studies would only be carried out in the 20th century.

0.4 PERCENT
THE PROPORTION OF **BABIES BORN AS IDENTICAL TWINS**



BRITISH NATURALIST Alfred Russel Wallace

(see 1855–58) published *The Geographical Distribution of Animals* in 1876. Co-founder of natural selection theory with Charles Darwin (see 1859–60), Wallace also made contributions to **biogeography**, including the concept of warning coloration in animals and the way barriers to hybridization, such as mountains, contribute to the evolution of new species.

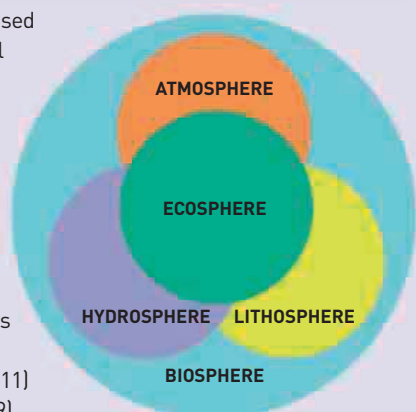
American inventors Elisha Gray (1835–1901) and **Alexander Graham Bell** (see panel, opposite) independently came up with the design of a working **telephone**. However, Bell won the race to patent the device on March 7, 1876. Three days later, he spoke the famous sentence: “Mr. Watson, come here, I want to see you.” into the machine, summoning his assistant from the next room.

Just about two months later, German inventor **Nikolaus Otto** (1832–91) completed building the first practical **four-stroke piston internal combustion engine** (see 1807–09).

In 1876, German neurologist **Karl Wernicke** (1848–1905) found that damage to a specific part of the brain—now called **Wernicke's area**—resulted in language disorders. This area of the brain is connected to Broca's area (see 1861–64) by a nerve

BIOSPHERE

The term biosphere is used to refer to the sum of all ecosystems. A closed, self-regulating system, the biosphere shows that living things do not exist in isolation from the nonliving world. Devised by Eduard Suess in 1875, the idea was developed by James Lovelock (b.1919) and Lynn Margulis (1938–2011) as Gaia theory (see 1979).



1874 Georg Cantor publishes his work on set theory

1874 Russian Sofia Kovalevskaya becomes the first woman to be awarded a doctorate in mathematics

1874 Othmar Zeidler synthesizes DDT

1874 William Stanley Jevons publishes *The Principles of Science: A Treatise on Logic and Scientific Method*

1875 Paul-Émile Lecoq de Boisbaudran discovers gallium

1875 Polish-German botanist Eduard Strasburger describes mitotic cell division in plants

1875 Austrian geologist Eduard Suess coins the term biosphere

1875 Francis Galton publishes his work on twin studies

1875 British chemist and physicist William Crookes develops a radiometer

1876 Alfred Russel Wallace publishes *The Geographical Distribution of Animals*

March 7, 1876 Alexander Graham Bell is granted a patent for the telephone